DisasterConnect - Project Scope & Implementation Plan

Project Overview

DisasterConnect is a disaster management platform built using React, Node.js, Express, MongoDB, AI services, and Progressive Web App technologies. The platform helps citizens, volunteers, rescue teams, NGOs, and administrators manage disaster situations efficiently.

Objectives

- Emergency SOS Management
- Shelter Management
- Missing Person Tracking
- Volunteer & Rescue Coordination
- Disaster Alerts & Notifications
- AI-Based Recommendations

Technology Stack

Frontend: React, TypeScript, Tailwind CSS, PWA
Backend: Node.js, Express.js, JWT Authentication
Database: MongoDB
AI Services: Python FastAPI, OpenAI APIs
Storage: AWS S3 / MinIO

Modules

1. Authentication & Authorization
2. User Management
3. Disaster Management
4. SOS Management
5. Shelter Management
6. Missing Person Tracking
7. Volunteer Management
8. Rescue Team Management
9. Resource Management
10. Alerts & Notifications
11. Analytics Dashboard
12. AI Assistant

MongoDB Collections

Users, Roles, Disasters, SOSAlerts, Shelters, Volunteers, RescueTeams, Resources, Messages, Alerts, MissingPersons, AuditLogs

System Architecture

React PWA -> Node.js API Gateway -> MongoDB -> AI Services -> Notification Service

Implementation Roadmap

Phase 1: Requirement Analysis & Design
Phase 2: Authentication & User Management
Phase 3: Disaster & SOS Modules
Phase 4: Shelter & Rescue Modules
Phase 5: Maps & Location Services
Phase 6: AI Integration
Phase 7: Analytics & Reports
Phase 8: Testing & Deployment

Expected Outcome

A scalable disaster-response platform capable of managing incidents, shelters, rescue operations, and emergency communications.